

MATH-334 Numerical Analysis

Newton's Divided Difference Interpolations

Dr. Yasir Ali (yali@ceme.nust.edu.pk)

DBS&H, CEME-NUST

April 25, 2020

Outline

① Topics Related to Interpolation Covered So Far

Curve Fitting, Regression and Interpolation

Why Polynomial Interpolation?

Polynomial Interpolation Problem

Lagrange Interpolating Polynomials

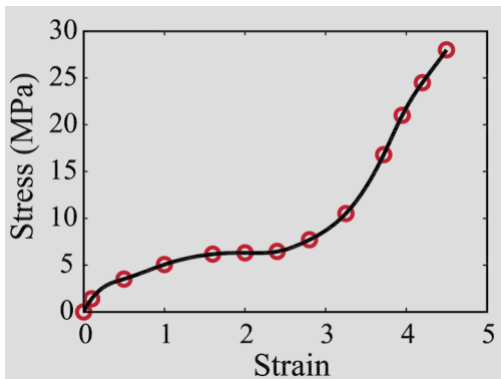
Derivation of Lagrange Int. for Two points

Derivation of Lagrange Int. for Three points

Derivation of Lagrange Int. for $n+1$ points

② Summary

Basic approach in **Interpolation** is to fit a curve or a series of curves that **pass directly through each of the points**



Polynomial Interpolation Problem

- Given**
- ① The $(n + 1)$ nodes: $\underbrace{x_0, x_1, \dots, x_n}_{n+1}$ **Data points**
 - ② The functional values

$f(x_0), f(x_1), \dots, f(x_n)$ on nodes $x_0, x_1, \dots, x_n$.

- ③ An Intermediate (non-tabulated) point: x^*

Predict $f(x^*)$: the value at $x = x^*$.

Polynomial Interpolation Problem

- Given**
- ① The $(n + 1)$ nodes: $\underbrace{x_0, x_1, \dots, x_n}_{n+1}$ **Data points**
 - ② The functional values

$f(x_0), f(x_1), \dots, f(x_n)$ on nodes $x_0, x_1, \dots, x_n$.

- ③ An Intermediate (non-tabulated) point: x^*

Predict $f(x^*)$: the value at $x = x^*$.

- ⊛ Polynomial interpolation consists of determining the unique polynomial $y = P(x)$ of order n that fits $n + 1$ data points.

Polynomial Interpolation Problem

- Given**
- ① The $(n + 1)$ nodes: $\underbrace{x_0, x_1, \dots, x_n}_{n+1 \text{ Data points}}$
 - ② The functional values

$f(x_0), f(x_1), \dots, f(x_n)$ on nodes $x_0, x_1, \dots, x_n$.

- ③ An Intermediate (non-tabulated) point: x^*

Predict $f(x^*)$: the value at $x = x^*$.

- ⊛ Polynomial interpolation consists of determining the unique polynomial $y = P(x)$ of order n that fits $n + 1$ data points.
- ⊛ This **polynomial provides a formula** to compute intermediate values.

- Lagrange Interpolating Polynomials

- Derivation of Lagrange Interpolating Polynomial for two points

To find a polynomial $y = P(x)$ of degree one that passes through the distinct points $(x_0, f(x_0))$ and $(x_1, f(x_1))$, using **Lagrange method**, we have to start with

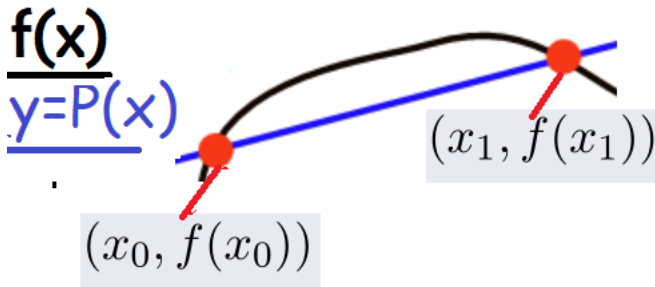
$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1), \quad \text{that satisfies}$$

To find a polynomial $y = P(x)$ of degree one that passes through the distinct points $(x_0, f(x_0))$ and $(x_1, f(x_1))$, using **Lagrange method**, we have to start with

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1), \quad \text{that satisfies}$$

(i) $P(x_0) = f(x_0),$

(ii) $P(x_1) = f(x_1),$

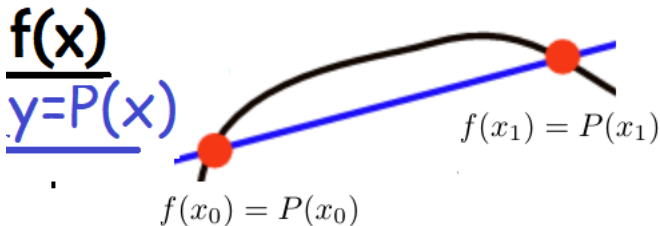


To find a polynomial $y = P(x)$ of degree one that passes through the distinct points $(x_0, f(x_0))$ and $(x_1, f(x_1))$, using **Lagrange method**, we have to start with

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1), \quad \text{that satisfies}$$

$$(i) \quad P(x_0) = f(x_0),$$

$$(ii) \quad P(x_1) = f(x_1),$$



To find a polynomial $y = P(x)$ of degree one that passes through the distinct points $(x_0, f(x_0))$ and $(x_1, f(x_1))$, using **Lagrange method**, we have to start with

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1), \quad \text{that satisfies}$$

$$(i) \quad P(x_0) = f(x_0),$$

$$(ii) \quad P(x_1) = f(x_1),$$



$$f(x_0) = P(x_0)$$

$$f(x_1) = P(x_1)$$

$$L_0(x_0) = 1$$

$$L_1(x_0) = 0$$

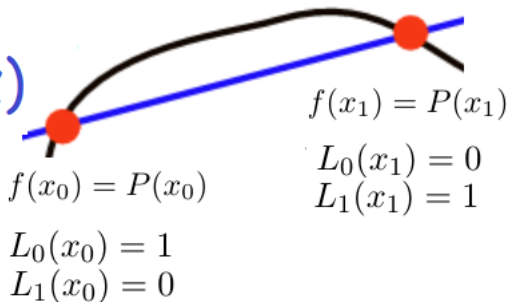
To find a polynomial $y = P(x)$ of degree one that passes through the distinct points $(x_0, f(x_0))$ and $(x_1, f(x_1))$, using **Lagrange method**, we have to start with

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1), \quad \text{that satisfies}$$

$$(i) \quad P(x_0) = f(x_0),$$

$$(ii) \quad P(x_1) = f(x_1),$$

$$\frac{f(x)}{y=P(x)}$$



Requirement for Lagrange Interpolating Polynomial

Requirement for Lagrange Interpolating Polynomial

To find a polynomial $y = P(x)$ of degree one that passes through the distinct points $(x_0, f(x_0))$ and $(x_1, f(x_1))$, using **Lagrange method**, we have to start with

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1), \quad \text{that satisfies}$$

(i) $P(x_0) = f(x_0)$, that is,

$$L_0(x_0) = 1 \quad \text{and} \quad L_1(x_0) = 0 \quad \text{at } x_0$$

(ii) $P(x_1) = f(x_1)$, that is,

$$L_0(x_1) = 0 \quad \text{and} \quad L_1(x_1) = 1 \quad \text{at } x_1$$

How can we define such $L_0(x)$ and $L_1(x)$?

- (i) $L_0(x_0) = 1$ *and* $L_1(x_0) = 0$ at x_0
- (ii) $L_0(x_1) = 0$ *and* $L_1(x_1) = 1$ at x_1

How can we define such $L_0(x)$ and $L_1(x)$?

$$(i) \quad L_0(x_0) = 1 \quad \text{and} \quad L_1(x_0) = 0 \quad \text{at } x_0$$

$$(ii) \quad L_0(x_1) = 0 \quad \text{and} \quad L_1(x_1) = 1 \quad \text{at } x_1$$

We can define

$$L_0(x) = x - x_1$$

How can we define such $L_0(x)$ and $L_1(x)$?

$$(i) \quad L_0(x_0) = 1 \quad \text{and} \quad L_1(x_0) = 0 \quad \text{at } x_0$$

$$(ii) \quad L_0(x_1) = 0 \quad \text{and} \quad L_1(x_1) = 1 \quad \text{at } x_1$$

We can define

$$L_0(x) = \frac{x - x_1}{x_0 - x_1}$$

How can we define such $L_0(x)$ and $L_1(x)$?

$$(i) \quad L_0(x_0) = 1 \quad \text{and} \quad L_1(x_0) = 0 \quad \text{at } x_0$$

$$(ii) \quad L_0(x_1) = 0 \quad \text{and} \quad L_1(x_1) = 1 \quad \text{at } x_1$$

We can define

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = \begin{cases} 1 & \text{if } x = x_0, \end{cases}$$

How can we define such $L_0(x)$ and $L_1(x)$?

$$(i) \quad L_0(x_0) = 1 \quad \text{and} \quad L_1(x_0) = 0 \quad \text{at } x_0$$

$$(ii) \quad L_0(x_1) = 0 \quad \text{and} \quad L_1(x_1) = 1 \quad \text{at } x_1$$

We can define

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = \begin{cases} 1 & \text{if } x = x_0, \\ 0 & \text{O.W} \end{cases}$$

How can we define such $L_0(x)$ and $L_1(x)$?

$$(i) \quad L_0(x_0) = 1 \quad \text{and} \quad L_1(x_0) = 0 \quad \text{at } x_0$$

$$(ii) \quad L_0(x_1) = 0 \quad \text{and} \quad L_1(x_1) = 1 \quad \text{at } x_1$$

We can define

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = \begin{cases} 1 & \text{if } x = x_0, \\ 0 & \text{O.W} \end{cases}, \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

How can we define such $L_0(x)$ and $L_1(x)$?

(i) $L_0(x_0) = 1$ and $L_1(x_0) = 0$ at x_0

(ii) $L_0(x_1) = 0$ and $L_1(x_1) = 1$ at x_1

We can define

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = \begin{cases} 1 & \text{if } x = x_0, \\ 0 & \text{O.W} \end{cases}, \quad L_1(x) = \frac{x - x_0}{x_1 - x_0} = \begin{cases} 1 & \text{if } x = x_1, \\ 0 & \text{O.W} \end{cases}$$

How can we define such $L_0(x)$ and $L_1(x)$?

$$(i) \quad L_0(x_0) = 1 \quad \text{and} \quad L_1(x_0) = 0 \quad \text{at } x_0$$

$$(ii) \quad L_0(x_1) = 0 \quad \text{and} \quad L_1(x_1) = 1 \quad \text{at } x_1$$

We can define

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = \begin{cases} 1 & \text{if } x = x_0, \\ 0 & \text{O.W} \end{cases}, \quad L_1(x) = \frac{x - x_0}{x_1 - x_0} = \begin{cases} 1 & \text{if } x = x_1, \\ 0 & \text{O.W} \end{cases}$$

The linear Lagrange interpolating polynomial through $(x_0, f(x_0))$ and $(x_1, f(x_1))$ is

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) = \frac{x - x_1}{x_0 - x_1}f(x_0) + \frac{x - x_0}{x_1 - x_0}f(x_1)$$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points $(2, 4)$ and $(5, 1)$.)

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

We have $x_0 = 2$, $x_1 = 5$ and $f(x_0) = 4$ $f(x_1) = 1$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

We have $x_0 = 2, \quad x_1 = 5$ and $f(x_0) = 4 \quad f(x_1) = 1$

$$L_0(x) = \frac{x - 5}{2 - 5}$$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

We have $x_0 = 2$, $x_1 = 5$ and $f(x_0) = 4$ $f(x_1) = 1$

$$L_0(x) = \frac{x - 5}{2 - 5}$$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

We have $x_0 = 2$, $x_1 = 5$ and $f(x_0) = 4$ $f(x_1) = 1$

$$L_0(x) = \frac{x - 5}{2 - 5} = \begin{cases} 1 & \text{if } x = 2, \end{cases}$$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

We have $x_0 = 2, \quad x_1 = 5$ and $f(x_0) = 4 \quad f(x_1) = 1$

$$L_0(x) = \frac{x - 5}{2 - 5} = \begin{cases} 1 & \text{if } x = 2, \\ 0 & \text{O.W} \end{cases}$$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

We have $x_0 = 2, \quad x_1 = 5$ and $f(x_0) = 4 \quad f(x_1) = 1$

$$L_0(x) = \frac{x - 5}{2 - 5} = \begin{cases} 1 & \text{if } x = 2, \\ 0 & \text{O.W} \end{cases} \quad L_1(x) = \frac{x - 2}{5 - 2}$$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

We have $x_0 = 2, \quad x_1 = 5$ and $f(x_0) = 4 \quad f(x_1) = 1$

$$L_0(x) = \frac{x - 5}{2 - 5} = \begin{cases} 1 & \text{if } x = 2, \\ 0 & \text{O.W} \end{cases}, \quad L_1(x) = \frac{x - 2}{5 - 2} = \begin{cases} 0 & \text{if } x = 2, \\ 1 & \text{O.W} \end{cases}$$

$$P(x) = \frac{-(x - 5)}{3}(4) + \frac{(x - 2)}{3}(1) = -x + 6$$

Example (Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).)

The Lagrange's Polynomial is $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$ where

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

We have $x_0 = 2, \quad x_1 = 5$ and $f(x_0) = 4 \quad f(x_1) = 1$

$$L_0(x) = \frac{x - 5}{2 - 5} = \begin{cases} 1 & \text{if } x = 2, \\ 0 & \text{O.W} \end{cases}, \quad L_1(x) = \frac{x - 2}{5 - 2} = \begin{cases} 0 & \text{if } x = 2, \\ 1 & \text{O.W} \end{cases}$$

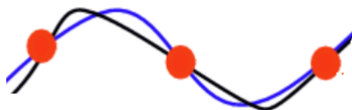
$$P(x) = \frac{-(x - 5)}{3}(4) + \frac{(x - 2)}{3}(1) = -x + 6 = \begin{cases} 4 & \text{if } x = 2, \\ 1 & \text{if } x = 5. \end{cases}$$

- Derivation of Lagrange Interpolating Polynomial for three points

Lagrange Interpolating Polynomial for Three Data Points

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2), \quad \text{that satisfies}$$

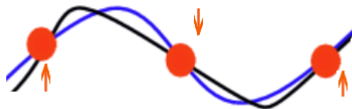
$$\begin{array}{l} f(x) \text{---} \\ y=p(x) \text{---} \end{array}$$



Lagrange Interpolating Polynomial for Three Data Points

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2), \quad \text{that satisfies}$$

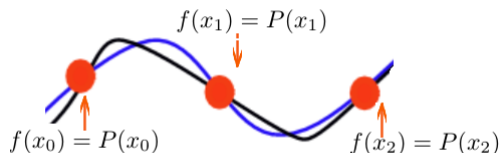
$$\begin{array}{l} f(x) \text{---} \\ y=p(X) \text{---} \end{array}$$



Lagrange Interpolating Polynomial for Three Data Points

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2), \quad \text{that satisfies}$$

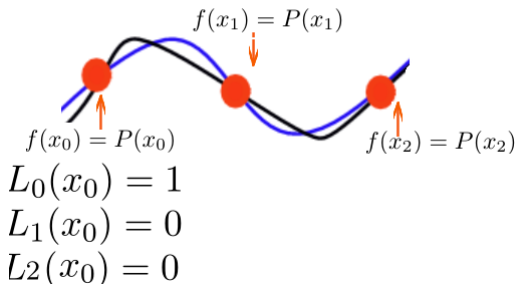
$$\begin{matrix} f(x) - \\ y = p(x) - \end{matrix}$$



Lagrange Interpolating Polynomial for Three Data Points

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2), \quad \text{that satisfies}$$

$$\begin{aligned} f(x) &= \\ y &= p(x) \end{aligned}$$



Lagrange Interpolating Polynomial for Three Data Points

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2), \quad \text{that satisfies}$$

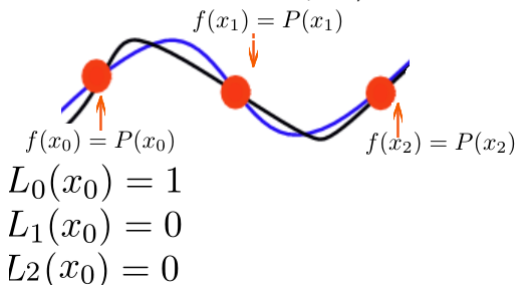
$f(x)$ —

$y=p(x)$ —

$$L_0(x_1) = 0$$

$$L_1(x_1) = 1$$

$$L_2(x_1) = 0$$



Lagrange Interpolating Polynomial for Three Data Points

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2), \quad \text{that satisfies}$$

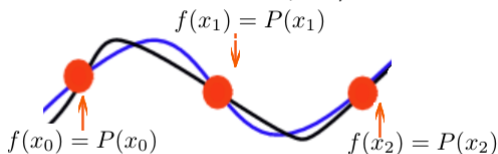
$f(x)$ —

$y=p(x)$ —

$$L_0(x_1) = 0$$

$$L_1(x_1) = 1$$

$$L_2(x_1) = 0$$



$$L_0(x_0) = 1$$

$$L_1(x_0) = 0$$

$$L_2(x_0) = 0$$

$$L_0(x_2) = 0$$

$$L_1(x_2) = 0$$

$$L_2(x_2) = 1$$

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	2.0	1/2
1	2.5	1/2.5
2	4	1/4

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	2.0	1/2
1	2.5	1/2.5
2	4	1/4

Nodes: $x_0 = 2.0$, $x_1 = 2.5$, $x_2 = 4$,
 $n = 2$

Functional values: $f_0 = 1/2$,
 $f_1 = 1/2.5$, $f_2 = 1/4$.

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	2.0	1/2
1	2.5	1/2.5
2	4	1/4

Nodes: $x_0 = 2.0$, $x_1 = 2.5$, $x_2 = 4$,
 $n = 2$

Functional values: $f_0 = 1/2$,
 $f_1 = 1/2.5$, $f_2 = 1/4$.

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$L_1(x) =$$

$$L_2(x) =$$

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	2.0	1/2
1	2.5	1/2.5
2	4	1/4

Nodes: $x_0 = 2, x_1 = 2.5, x_2 = 4,$
 $n = 2$

Functional values: $f_0 = 1/2,$
 $f_1 = 1/2.5, f_2 = 1/4.$

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)}$$

$$L_1(x) =$$

$$L_2(x) =$$

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	2.0	1/2
1	2.5	1/2.5
2	4	1/4

Nodes: $x_0 = 2.0$, $x_1 = 2.5$, $x_2 = 4$,
 $n = 2$

Functional values: $f_0 = 1/2$,
 $f_1 = 1/2.5$, $f_2 = 1/4$.

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

$$L_1(x) =$$

$$L_2(x) =$$

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	2.0	1/2
1	2.5	1/2.5
2	4	1/4

Nodes: $x_0 = 0$, $x_1 = 2.5$, $x_2 = 4$,
 $n = 2$

Functional values: $f_0 = 1/2$,
 $f_1 = 1/2.5$, $f_2 = 1/4$.

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 2)(x - 4)}{(0.5)(-1.5)}$$

$$L_2(x) =$$

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	2.0	1/2
1	2.5	1/2.5
2	4	1/4

Nodes: $x_0 = 2.0$, $x_1 = 2.5$, $x_2 = 4$,
 $n = 2$

Functional values: $f_0 = 1/2$,
 $f_1 = 1/2.5$, $f_2 = 1/4$.

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 2)(x - 4)}{(0.5)(-1.5)} = -\frac{1}{0.75}(x - 2)(x - 4) = \frac{x^2 - 6x + 8}{0.75}$$

$$L_2(x) =$$

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	2.0	1/2
1	2.5	1/2.5
2	4	1/4

Nodes: $x_0 = 2.0$, $x_1 = 2.5$, $x_2 = 4$,
 $n = 2$

Functional values: $f_0 = 1/2$,
 $f_1 = 1/2.5$, $f_2 = 1/4$.

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 2)(x - 4)}{(0.5)(-1.5)} = -\frac{1}{0.75}(x - 2)(x - 4) = \frac{x^2 - 6x + 8}{0.75}$$

$$L_2(x) = \frac{(x - x_1)(x - x_0)}{(x_2 - x_1)(x_2 - x_0)} = \frac{1}{3}(x - 2.5)(x - 2) = \frac{x^2 - 4.5x + 5}{3}$$

x_i	2.0	2.5	4
$f(x_i)$	1/2	1/2.5	1/4

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 2)(x - 4)}{(0.5)(-1.5)} = -\frac{1}{0.75}(x - 2)(x - 4) = \frac{x^2 - 6x + 8}{0.75}$$

$$L_2(x) = \frac{(x - x_1)(x - x_0)}{(x_2 - x_1)(x_2 - x_0)} = \frac{1}{3}(x - 2.5)(x - 2) = \frac{x^2 - 4.5x + 5}{3}$$

$$L_0(x) = x^2 - 6.5x + 10, \quad L_1(x) = \frac{x^2 - 6x + 8}{0.75} \text{ and } L_2(x) = \frac{x^2 - 4.5x + 5}{3}$$

x_i	2.0	2.5	4
$f(x_i)$	1/2	1/2.5	1/4

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 2)(x - 4)}{(0.5)(-1.5)} = -\frac{1}{0.75}(x - 2)(x - 4) = \frac{x^2 - 6x + 8}{0.75}$$

$$L_2(x) = \frac{(x - x_1)(x - x_0)}{(x_2 - x_1)(x_2 - x_0)} = \frac{1}{3}(x - 2.5)(x - 2) = \frac{x^2 - 4.5x + 5}{3}$$

$$L_0(x) = x^2 - 6.5x + 10, \quad L_1(x) = \frac{x^2 - 6x + 8}{0.75} \text{ and } L_2(x) = \frac{x^2 - 4.5x + 5}{3}$$

Step 2. Find the interpolating polynomial $P_2(x)$:

$$P_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) = \frac{1}{2}L_0(x) + \frac{1}{2.5}L_1(x) + \frac{1}{4}L_2(x)$$

x_i	2.0	2.5	4
$f(x_i)$	1/2	1/2.5	1/4

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 2)(x - 4)}{(0.5)(-1.5)} = -\frac{1}{0.75}(x - 2)(x - 4) = \frac{x^2 - 6x + 8}{0.75}$$

$$L_2(x) = \frac{(x - x_1)(x - x_0)}{(x_2 - x_1)(x_2 - x_0)} = \frac{1}{3}(x - 2.5)(x - 2) = \frac{x^2 - 4.5x + 5}{3}$$

$$L_0(x) = x^2 - 6.5x + 10, \quad L_1(x) = \frac{x^2 - 6x + 8}{0.75} \text{ and } L_2(x) = \frac{x^2 - 4.5x + 5}{3}$$

Step 2. Find the interpolating polynomial $P_2(x)$:

$$\begin{aligned} P_2(x) &= f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) = \frac{1}{2}L_0(x) + \frac{1}{2.5}L_1(x) + \frac{1}{4}L_2(x) \\ &= \frac{1}{2}(x^2 - 6.5x + 10) + \frac{1}{2.5} \frac{(x^2 - 6x + 8)}{0.75} + \frac{1}{4} \frac{(x^2 - 4.5x + 5)}{3} \end{aligned}$$

x_i	2	2.5	4
$f(x_i)$	1/2	1/2.5	1/4

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2)$$

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

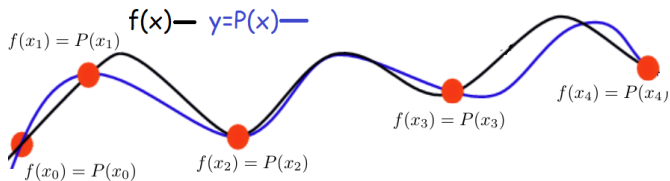
$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 2)(x - 4)}{(0.5)(-1.5)} = -\frac{1}{0.75}(x - 2)(x - 4) = \frac{x^2 - 6x + 8}{0.75}$$

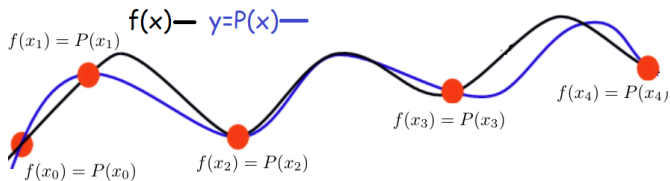
$$L_2(x) = \frac{(x - x_1)(x - x_0)}{(x_2 - x_1)(x_2 - x_0)} = \frac{1}{3}(x - 2.5)(x - 2) = \frac{x^2 - 4.5x + 5}{3}$$

$$L_0(x) = x^2 - 6.5x + 10, \quad L_1(x) = \frac{x^2 - 6x + 8}{0.75} \text{ and } L_2(x) = \frac{x^2 - 4.5x + 5}{3}$$

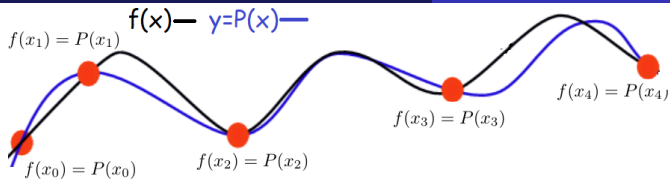
Step 2. Find the interpolating polynomial $P_2(x)$:

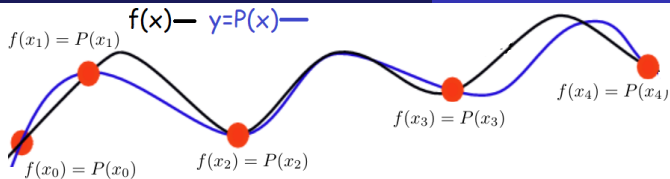
$$\begin{aligned} P_2(x) &= f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) = \frac{1}{2}L_0(x) + \frac{1}{2.5}L_1(x) + \frac{1}{4}L_2(x) \\ &= \frac{1}{2}(x^2 - 6.5x + 10) + \frac{1}{2.5} \frac{(x^2 - 6x + 8)}{0.75} + \frac{1}{4} \frac{(x^2 - 4.5x + 5)}{3} \quad \boxed{P_2(3) = 0.3250.} \end{aligned}$$



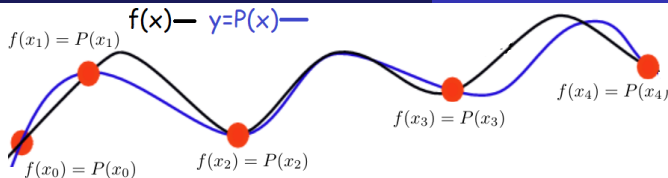


- Derivation of Lagrange Interpolating Polynomial for $n+1$ points





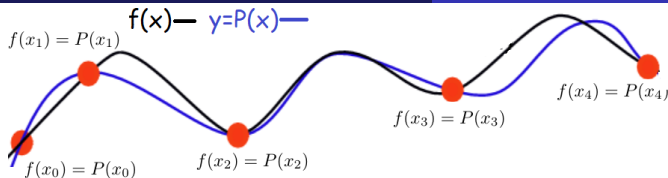
$$L_k(x_i) = L_{n,k}(x_i) = \begin{cases} 1 & \text{if } k = i, \\ 0 & \text{otherwise.} \end{cases}$$



$$L_k(x_i) = L_{n,k}(x_i) = \begin{cases} 1 & \text{if } k = i, \\ 0 & \text{otherwise.} \end{cases}$$

To satisfy $L_k(x_i) = 0$ for each $i \neq k$ requires that the numerator of $L_k(x)$ contain the term

$$\underbrace{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}_{\text{product of all factors except } x_k}.$$

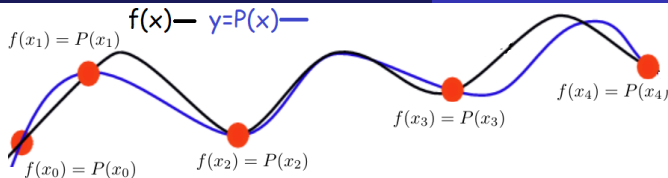


$$L_k(x_i) = L_{n,k}(x_i) = \begin{cases} 1 & \text{if } k = i, \\ 0 & \text{otherwise.} \end{cases}$$

To satisfy $L_k(x_i) = 0$ for each $i \neq k$ requires that the numerator of $L_k(x)$ contain the term

$$\underbrace{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}_{\text{product of all factors except } x_k}.$$

To satisfy $L_k(x_k) = 1$, the denominator of $L_k(x)$ must be this same term but evaluated at $x = x_k$. Thus



$$L_k(x_i) = L_{n,k}(x_i) = \begin{cases} 1 & \text{if } k = i, \\ 0 & \text{otherwise.} \end{cases}$$

To satisfy $L_k(x_i) = 0$ for each $i \neq k$ requires that the numerator of $L_k(x)$ contain the term

$$\underbrace{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}_{\text{product of all factors except } x_k}.$$

To satisfy $L_k(x_k) = 1$, the denominator of $L_k(x)$ must be this same term but evaluated at $x = x_k$. Thus

$$L_k(x) = \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}$$

If $x_0, x_1, \dots, x_n$ are $n+1$ distinct numbers and f is a function whose values are given at these numbers, then a unique polynomial $P(x)$ of degree at most n exists with

$$f(x_k) = P(x_k), \quad \text{for each } k = 0, 1, \dots, n.$$

This polynomial is given by

$$\begin{aligned} P(x) &= L_0(x)f(x_0) + L_1(x)f(x_1) + \dots + L_n(x)f(x_n) \\ &= \sum_{k=0}^n f(x_k)L_k(x), \end{aligned}$$

where, for each $k = 0, 1, \dots, n$,

$$\begin{aligned} L_k(x) &= \frac{(x-x_0)(x-x_1)\cdots(x-x_{k-1})(x-x_{k+1})\cdots(x-x_n)}{(x_k-x_0)(x_k-x_1)\cdots(x_k-x_{k-1})(x_k-x_{k+1})\cdots(x_k-x_n)} \\ &= \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x-x_i}{x_k-x_i} \end{aligned}$$

$$\begin{aligned}L_k(x) &= \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)} \\&= \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i}\end{aligned}$$

$$\begin{aligned}L_k(x) &= \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)} \\&= \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i}\end{aligned}$$

It can easily be seen that

$$L_k(x_i) = \begin{cases} 0, & \text{when } i \neq k; \\ 1, & \text{otherwise.} \end{cases}$$

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	0	7
1	1	13
2	2	21
3	4	43

Nodes: $x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 4, n = 3$

Functional values: $f_0 = 7, f_1 = 13, f_2 = 21, f_3 = 43$.

To find the polynomial we will use the following formula

$$P_3(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)$$

Explicitly construct the Lagrange Interpolating Polynomial and interpolate $f(3)$.

i	x_i	$y_i = f(x_i) = f_i$
0	0	7
1	1	13
2	2	21
3	4	43

Nodes: $x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 4, n = 3$

Functional values: $f_0 = 7, f_1 = 13, f_2 = 21, f_3 = 43$.

To find the polynomial we will use the following formula

$$P_3(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)$$

$$P_3(x) = 7L_0(x) + 13L_1(x) + 21L_2(x) + 43L_3(x)$$

$$L_0(x) = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} = \frac{(x - 1)(x - 2)(x - 4)}{(0 - 1)(0 - 2)(0 - 4)}$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)} = \frac{(x)(x - 2)(x - 4)}{(1)(1 - 2)(1 - 4)}$$

$$L_2(x) = \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} = \frac{(x)(x - 1)(x - 4)}{(2)(2 - 1)(2 - 4)}$$

To find the polynomial we will use the following formula

$$P_3(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)$$

To find the polynomial we will use the following formula

$$P_3(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)$$

$$P_3(x) = 7L_0(x) + 13L_1(x) + 21L_2(x) + 43L_3(x)$$

$$L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x-1)(x-2)(x-4)}{(0-1)(0-2)(0-4)}$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x)(x-2)(x-4)}{(1)(1-2)(1-4)}$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x)(x-1)(x-4)}{(2)(2-1)(2-4)}$$

$$L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x)(x-1)(x-2)}{(4)(4-1)(4-2)}$$

To find the polynomial we will use the following formula

$$P_3(x) = \textcolor{red}{7}L_0(x) + \textcolor{blue}{13}L_1(x) + \textcolor{green}{21}L_2(x) + \textcolor{black}{43}L_3(x)$$

$$\textcolor{red}{L_0(x)} = \frac{\textcolor{red}{(x-1)(x-2)(x-4)}}{\textcolor{red}{-8}}$$

$$\textcolor{green}{L_2(x)} = \frac{\textcolor{green}{(x)(x-1)(x-4)}}{\textcolor{green}{-4}}$$

$$\textcolor{blue}{L_1(x)} = \frac{\textcolor{blue}{(x)(x-2)(x-4)}}{\textcolor{blue}{3}}$$

$$L_3(x) = \frac{(x)(x-1)(x-2)}{24}$$

To find the polynomial we will use the following formula

$$P_3(x) = 7L_0(x) + 13L_1(x) + 21L_2(x) + 43L_3(x)$$

$$L_0(x) = \frac{(x-1)(x-2)(x-4)}{-8}$$

$$L_1(x) = \frac{(x)(x-2)(x-4)}{3}$$

$$L_2(x) = \frac{(x)(x-1)(x-4)}{-4}$$

$$L_3(x) = \frac{(x)(x-1)(x-2)}{24}$$

For $P_3(x)$ at $x = 3$

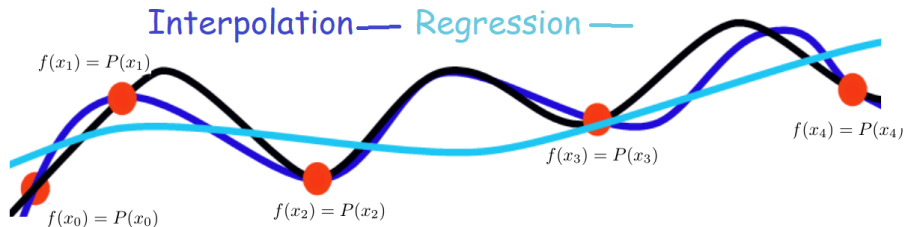
$$L_0(3) = \frac{(3-1)(3-2)(-4)}{-8} = \frac{1}{4}, \quad L_1(3) = \frac{(3)(3-2)(3-4)}{3} = -1$$

$$L_2(3) = \frac{(3)(3-1)(3-4)}{-4} = \frac{3}{2}, \quad L_3(3) = \frac{(3)(3-1)(3-2)}{24} = \frac{1}{4}$$

Thus $P_3(3) = 7(\frac{1}{4}) + 13(-1) + 21(\frac{3}{2}) + 43(\frac{1}{4}) = 31$.

Summary

Interpolation & Regression:



Summary

Lagrange Interpolation Polynomial:

$$\begin{aligned} P(x) &= L_{n,0}(x)f(x_0) + L_{n,1}(x)f(x_1) + \cdots + L_{n,n}(x)f(x_n) \\ &= \sum_{k=0}^n f(x_k)L_k(x), \text{ where} \end{aligned}$$

$$\begin{aligned} L_k(x) &= \frac{(x-x_0)(x-x_1)\cdots(x-x_{k-1})(x-x_{k+1})\cdots(x-x_n)}{(x_k-x_0)(x_k-x_1)\cdots(x_k-x_{k-1})(x_k-x_{k+1})\cdots(x_k-x_n)} \\ &= \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x-x_i}{x_k-x_i} \end{aligned}$$

This gives

$$P(x_i) = f(x_i) \quad \text{for all } i = 0, 1, 2, \dots, n.$$

Way forward

- Newton's Divided Difference Interpolation solved example
- Forward/Backward Difference Interpolation
- Equally Spaced Data